But, a circular slit (aperture) produces diffraction pattern of concentric circles as shown in Figure 7.23. These are known as Airy's discs. Most of the optical instruments form images of objects only through the circular slits. The condition for central maximum (or) first minimum for circular slit is,

$$a \sin\theta = 1.22\,\lambda \qquad (7.52)$$

Here, the numerical value 1.22 appears in the expression for central maximum (or) first minimum formed by circular slits. This involves higher level mathematics that is not shown here.

W9N9D3

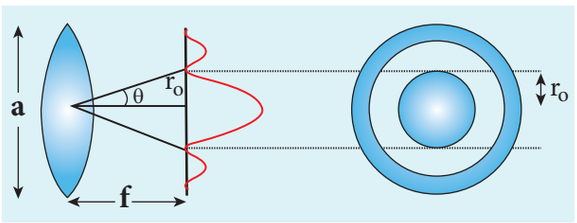


Figure 7.23 Airy's discs

For small angles, sin $\theta \approx \theta$, the above equation becomes,

$$a\,\theta = 1.22\,\lambda$$

Rewriting further,

$$\theta = \frac{1.22\lambda}{a} \qquad (7.53)$$

Form the geometry, $\theta = \frac{r_0}{f}$

Substituting for θ in equation (7.53) and rearranging gives

$$r_0 = \frac{1.22\lambda f}{a} \qquad (7.54)$$

For example, let two point-sources of light close to each other form image on a screen. The diffraction pattern of one point-source may overlap with another and produce a blurred image (or) un-resolved image as shown in Figure 7.24(a). To obtain a quality image (or) well resolved image, the two point-sources must be kept apart in such a way that their diffraction patterns do not overlap as shown in Figure 7.24(c).

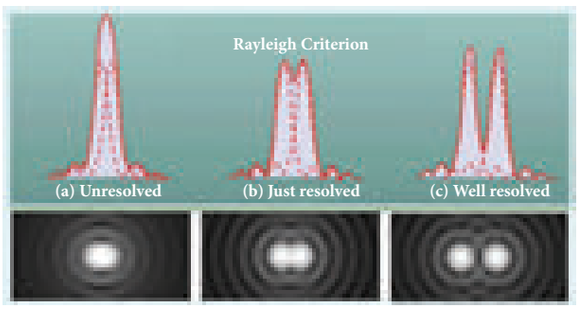


Figure 7.24 Rayleigh's criterion

According to *Rayleigh's criterion*, the two points on an image are said to be just resolved when the central maximum of one diffraction pattern coincides with the first minimum of the other and vice-versa as shown in Figure 7.24(b). In other words, the distance between the two central maxima must be at least r_o. Hence, r_o is called ***spatial resolution*** given by the equation (7.54) and the corresponding θ is said to be ***angular resolution*** given by the equation (7.53) respectively. It shows that for better resolution, the wavelength of light used must be as small as possible and the size of the aperture of the instrument must be as large as possible.

The ability of an optical instrument to distinguish the two closely adjacent objects (or) two points on the same object is said to be the resolving power of the instrument. In general, the term resolution is pertaining to the quality of

UNIT-7(XII-Physics_Vol-2)_new.indd 78 21-07-2020 16:36:14